Single-cell phylogenomics from scRNA-seq SNPs using scPhylogenomics: an end-to-end workflow for clonal tumor evolution analysis

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Editorial summary

We present **scPhylogenomics**, a modular and containerized end-to-end workflow that infers clonal tumor evolution directly from 10x Genomics single-cell RNA-sequencing (scRNA-seq) data. The protocol exploits somatic single-nucleotide variants (SNVs) within transcriptomic reads to reconstruct high-resolution clonal histories without requiring matched single-cell DNA sequencing.

Abstract

We present scPhylogenomics, a modular and containerized computational workflow designed to infer clonal tumor evolution directly from 10x Genomics scRNA-seq data. Unlike standard transcriptomic pipelines that focus solely on gene expression-based cell states, this protocol exploits somatic single-nucleotide variants (SNVs) within transcriptomic reads to reconstruct high-resolution evolutionary histories. The procedure is organized into five automated modules: rigorous preprocessing (SoupX, DoubletFinder, and miQC), reference-assisted cell typing (Seurat and SingleR), ploidy-based malignancy stratification (CopyKAT), optimized single-cell SNV calling (cellsnp-lite), and maximum-likelihood phylogeny reconstruction (FastTree or IQ-TREE). A key advantage of

scPhylogenomics over existing tools is its unified interface for clonal clustering, offering users a choice between deep-learning–based manifold methods (SNPmanifold) or hybrid hierarchical weighted non-negative matrix factorization (WNMF) to resolve complex subclonal architectures. Using multi-site samples from the MSK SPECTRUM HGSOC study as a primary case study, users can expect to distinguish malignant from non-malignant cells, identify site-enriched clones, and generate annotated phylogenetic trees that reflect spatial patterns of metastasis. The end-to-end execution of the workflow typically takes 24–48 hours depending on dataset size and available high-performance computing (HPC) resources, requiring basic experience with command-line and containerized environments.

Key references

- Workflow and case study application — *Wafula et al., in preparation*.
- 10x Genomics Cell Ranger pipeline — `@url:https://www.10xgenomics.com/support/software/cell-ranger`.
- SoupX, DoubletFinder, miQC — preprocessing references (TODO: add DOIs).
- Seurat and SingleR — cell-typing references (TODO: add DOIs).
- CopyKAT — ploidy inference reference (TODO: add DOI).
- cellsnp-lite — single-cell genotyping reference (TODO: add DOI).
- FastTree, IQ-TREE — maximum-likelihood phylogenetics references (TODO: add DOIs).
- SNPmanifold — VAE-based clonal clustering reference (TODO: add DOI).

Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed cancer research by enabling the resolution of intratumoral transcriptional heterogeneity at single-cell resolution. Conventional scRNA-seq analysis workflows, however, focus predominantly on gene expression–based questions — clustering cells into phenotypic states, inferring trajectories, and identifying differentially expressed genes — while largely overlooking the somatic single-nucleotide variants (SNVs) embedded in expressed transcripts. In contrast, the reconstruction of tumor clonal architecture and evolutionary histories is typically pursued using bulk or single-cell DNA sequencing (scDNA-seq) and dedicated tumor phylogenetics tools. As a result, most scRNA-seq datasets remain underexploited with respect to clonal evolution, despite containing sequence-level information that could, in principle, support phylogenetic inference.

Inferring clonal structure from scRNA-seq is challenging. Coverage is restricted to expressed loci and shows strong locus- and cell-specific dropout, 3′/5′ capture biases in droplet-based platforms distort the representation of coding regions, and amplification noise and allelic imbalance can mimic homoplasy or back-mutation. These properties make scRNA-seq intrinsically less suited to *de novo* variant discovery than scDNA-seq, but they do not preclude phylogenomics if models and workflows explicitly account for transcriptomic noise and the targeting of expressed SNVs. Existing tumor phylogenetics methods such as SCITE, SCARLET and related frameworks were primarily developed for bulk or scDNA-seq data and assume high-quality, genome-wide variant calls with well-characterized error models. Moreover, there is currently no widely adopted end-to-end protocol that takes users from raw scRNA-seq data through ambient RNA and doublet removal, cell-type annotation, ploidy-based tumor/normal stratification, scRNA-seq–tailored SNV calling, generation of SNP-based multiple sequence alignments (MSAs) and clonal clustering, to produce interpretable clonal phylogenies.

Here we present **scPhylogenomics**, a fully modular, scalable, and reproducible computational workflow that performs single-cell phylogenomics from 10x Genomics Chromium scRNA-seq data by integrating RNA-based quality control, reference-guided cell-type annotation, copy-number–based ploidy inference and SNV-based phylogeny reconstruction into a single protocol. The workflow is implemented as a GitHub repository with Docker and Singularity container images, and we illustrate its use on high-grade serous ovarian cancer (HGSOC) samples from the MSK SPECTRUM study (SPECTRUM-OV-003).

Development of the protocol

The scPhylogenomics protocol builds on three conceptual strands of prior work: (i) robust preprocessing and quality control for droplet-based scRNA-seq, (ii) reference-based cell-type annotation and ploidy inference from expression data, and (iii) SNV-based tumor phylogenetics and clonal clustering.

First, the preprocessing strategy (see *Procedure*, Module 1) integrates established tools for ambient RNA correction (SoupX), doublet detection (DoubletFinder) and quality control based on mitochondrial content and gene detection (miQC). These tools have been widely used in the scRNA-seq community but are usually deployed piecemeal, with users needing to embed them in bespoke pipelines. scPhylogenomics formalizes their combination into a standardized module that can be reused across projects.

Second, the protocol leverages reference-based methods to annotate cell types and infer ploidy. For cell-type annotation, scPhylogenomics uses Seurat for integration and clustering and SingleR (with celldex references or user-supplied atlases) to transfer labels from well-characterized reference datasets to new scRNA-seq data (see *Procedure*, Module 2). For ploidy inference, the protocol employs the **copykat** R package, which infers genome-wide copy-number profiles from expression data and classifies cells as aneuploid (putative malignant) or diploid (non-malignant), thereby operationalizing the observation that many solid tumors exhibit pervasive aneuploidy while stromal and immune cells remain near-diploid (see *Procedure*, Module 3).

Third, scPhylogenomics extends ideas from tumor phylogenetics and single-cell genotyping to scRNA-seq. It adopts **cellsnp-lite**, originally developed for efficient genotyping of known SNPs from scRNA-seq and scDNA-seq, to perform cell-type–specific SNV calling from 10x Chromium BAM files and to generate genotype matrices (allele depth, total depth and other alleles) (see *Procedure*, Module 4, scripts `02-snv-calling.py` and `utils/run-cellsnp-lite.sh`). From these matrices, the protocol constructs SNP-based MSAs in which each cell is encoded as a nucleotide sequence across ordered SNP loci, together with binary/tertiary mutation profiles (0/1/3 for reference/alternate/missing) for clustering (see `03-generate-snp-msa.py`). This representation allows the use of maximum-likelihood phylogenetic software such as FastTree and IQ-TREE and supports downstream clonal clustering using both a deep-learning variational autoencoder (SNPmanifold) and a hierarchical weighted non-negative matrix factorization (WNMF) approach (see *Procedure*, Module 5, `01-phylogeny-inference.py` and `03-snp-clustering.py`).

The development of scPhylogenomics was motivated by the need to resolve clonal divergence in tumor scRNA-seq data across both complex multi-site cohorts, such as the MSK SPECTRUM HGSOC study, and more conventional single-site or single-sample datasets, where pervasive aneuploidy and intra-tumoral heterogeneity pose similar analytical challenges. The protocol formalizes an analysis workflow we developed, organizing the method into five modules that can be run sequentially as an end-to-end pipeline or selectively applied to address related questions in tumor evolution and clonal architecture.

Overview of the procedure

The scPhylogenomics workflow is organized into five continuous modules that transform Cell Ranger–processed 10x scRNA-seq data into annotated clonal phylogenies (Fig. {@fig:workflow-overview}).

1. **Data Preprocessing.** This module (*Procedure*, Module 1) automates high-throughput quality control of 10x Genomics Chromium scRNA-seq data by sequentially removing ambient RNA, excluding doublets, and filtering low-quality cells to generate cleaned feature-barcode matrices for each sample. Starting from the raw Cell Ranger count outputs (barcodes, fea-

tures, and count matrices), the wrapper script `run-data-preprocessing.sh` calls three R scripts in series: `01-remove-ambient-rna.R` (SoupX), `02-remove-doublets.R` (DoubletFinder), and `03-remove-compromised.R` (miQC).

2. **Cell Typing.** This module (*Procedure*, Module 2) consolidates miQC-filtered samples into a merged or integrated Seurat object and performs reference-assisted cell-type annotation. The wrapper `run-cell-typing.sh` runs `01-consolidate-samples.R` (Seurat, Harmony/CCA/FastMNN/RPCA) and `02-annotate-cell-types.R` (SingleR with celldex, custom Seurat objects, or external mapping files) and produces UMAPs, heatmaps, and annotation tables.

3. **Ploidy Inference.** This module (*Procedure*, Module 3) applies `01-classify-cell-ploidy.R` to annotated Seurat objects using **copykat** to infer genome-wide copy-number profiles and stratify cells into aneuploid (malignant), diploid (normal), or uncertain classes.

4. **SNV Calling.** This module (*Procedure*, Module 4) performs cell type–specific somatic SNV calling and constructs SNP-based pseudo-MSAs using `01-get-cell-type-barcodes.R`, `02-snv-calling.py` (cellsnp-lite driver), and `03-generate-snp-msa.py`.

5. **Phylogeny Inference.** This module (*Procedure*, Module 5) reconstructs single-cell clonal histories by coupling maximum-likelihood tree building (`01-phylogeny-inference.py` — FastTree/IQ-TREE) with dual clonal clustering strategies (`03-snp-clustering.py` — SNPmanifold or hierarchical WNMF) and overlays clone, ploidy, cell type, and sample metadata via `04-infer-clonal-phylogeny.R`.

A schematic flowchart (Fig. {@fig:workflow-overview}) summarizes the data flow and outputs of each module, and the *Procedure* section describes the execution of these scripts in detail.

Schematic of the scPhylogenomics end-to-end workflow showing the five modules — data preprocessing, cell typing, ploidy inference, SNV calling and phylogeny inference — together with the principal inputs and outputs of each module. Placeholder figure — replace with the final content/images/workflow-overview.png exported from the lab figure master.

Schematic of the scPhylogenomics end-to-end workflow showing the five modules — data preprocessing, cell typing, ploidy inference, SNV calling and phylogeny inference — together with the principal inputs and outputs of each module. **Placeholder figure — replace with the final `content/images/workflow-overview.png` exported from the lab figure master.**

Applications

scPhylogenomics is intended for researchers who wish to leverage scRNA-seq data to study clonal tumor evolution, particularly in settings where matched scDNA-seq is unavailable or impractical. Proven and potential applications include:

- **Reconstruction of tumor clonal phylogenies from scRNA-seq.** By generating SNP-based MSAs and mutation profiles from scRNA-seq, scPhylogenomics enables maximum-likelihood tree inference and clonal clustering at single-cell resolution.
- **Dissection of intratumoral heterogeneity and metastatic spread.** In the MSK SPECTRUM HGSOC case study, the protocol distinguishes malignant from non-malignant cells, reveals aneuploid clones enriched at specific intraperitoneal sites and reconstructs phylogenies consistent with spatial patterns of metastasis.
- **Integration of genotype and transcriptional phenotype.** Because clonal assignments and ploidy labels can be mapped back onto expression-derived embeddings (e.g., UMAPs), scPhylogenomics supports joint analysis of evolutionary lineage, cell state and microenvironmental context.
- **Comparative studies across tumor types.** The workflow is demonstrated on HGSOC, TNBC and AML datasets to illustrate its generalizability across epithelial and hematologic malignancies.
- **Method benchmarking and development.** The modular design (e.g., pluggable SNV filters, clustering methods) makes scPhylogenomics suitable as a testbed for new scRNA-seq genotyping models, phylogenetic algorithms or integrative multi-omic approaches.

Comparison to other methods

Several existing methods address individual components of the scPhylogenomics workflow, but to our knowledge no published protocol provides an integrated, containerized pipeline for scRNA-seq–based tumor phylogenomics. On the preprocessing and quality-control side, frameworks such as Seurat, Scanpy and scater implement normalization, dimension reduction and clustering but leave ambient RNA, doublet detection and mitochondrial QC to separate tools. SoupX, DoubletFinder and miQC are widely used to address these issues, yet users typically need to orchestrate them manually. scPhylogenomics standardizes this combination and ensures that barcodes propagated into SNV calling are already filtered for major technical artifacts.

For SNV calling from scRNA-seq, general-purpose tools (for example, GATK RNA-seq best practices) are not optimized for ultra-sparse single-cell data and may produce high false-positive rates. **cellsnp-lite** was designed specifically for efficient genotyping of biallelic SNPs from scRNA-seq or scDNA-seq, providing fast pileup and genotyping of either detected or user-supplied common SNPs and outputting sparse SNP-by-cell count and genotype matrices. Alternative tools such as scSplit, demuxlet and vireo focus on donor demultiplexing and genotype assignment rather than phylogenetically informative SNV discovery. Building on cellsnp-lite, scPhylogenomics adds optional filtering against common SNP panels, RNA-editing catalogs and Panels of Normals, and generates SNP-based MSAs plus binary/tertiary mutation matrices required for downstream phylogenetic and clustering analyses.

In tumor phylogenetics, methods such as SCITE, SCARLET, SCClone and SECEDO were developed primarily for scDNA-seq or bulk data and model clonal evolution using binary genotype matrices under infinite-sites or finite-sites assumptions. They offer sophisticated probabilistic treatment of sequencing errors and copy-number events, but generally assume precomputed SNV calls and do not integrate scRNA-seq–specific preprocessing or the construction of nucleotide MSAs. Moreover, they often treat SNVs as abstract binary characters rather than sequences, limiting direct use of general phylogenetic software such as FastTree, IQ-TREE or RAxML. scPhylogenomics explicitly generates SNP-based MSAs, enabling users to apply standard molecular phylogenetic models to scRNA-seq–derived SNVs while also providing clonal clustering on mutation profiles.

Several end-to-end scRNA-seq pipelines (e.g., nf-core/scrnaseq, scdrake and other standardized workflows) emphasize expression-based analyses — QC, normalization, clustering and differential expression — but do not implement SNV calling or phylogeny reconstruction from scRNA-seq. Conversely, existing tumor phylogenomics workflows have focused on DNA-based input and typically lack modules for scRNA-seq–specific QC, cell-type annotation and ploidy detection.

The main advantages of scPhylogenomics relative to these alternatives are (i) its explicit focus on SNV-based phylogenomics from scRNA-seq, (ii) the integration of ploidy-based tumor/normal stratification to restrict SNV calling to malignant cells, (iii) the generation of both SNP MSAs and mutation profile matrices, and (iv) containerized implementation (Docker/Singularity) that facilitates reproducible deployment on workstations and HPC clusters. Its main limitations, discussed in detail elsewhere in the manuscript, include reliance on expressed loci, dependence on aneuploidy-based tumor classification, and computational cost for very large datasets.

Experimental design

Considerations before starting

Several aspects of experimental design should be considered before implementing scPhylogenomics:

- **Data type and sequencing platform.** The protocol is optimized for 10x Chromium 3′/5′ scRNA-seq data processed by Cell Ranger, where gene expression matrices and BAM files with cell barcodes and UMIs are readily available. Application to other platforms (e.g., Drop-seq, inDrop, Smart-seq) is feasible in principle but may require adaptation of input formats and adjustment of SNV filtering thresholds.
- **Tumor type and aneuploidy burden.** Because ploidy inference with copykat is only one (optional) route for tumor/normal stratification in scPhylogenomics, the workflow is especially effective when malignant cells exhibit clear genome-wide aneuploidy, such as in many HGSOC, TNBC, and other chromosomally unstable solid tumors. SNV calling is not restricted to copykat-defined tumor cells; it can be performed based on user-specified cell types deemed malignant from external information (e.g., marker expression or prior annotations), which is particularly useful in near-diploid cancers (e.g., many pediatric or hematologic malignancies) where copy-number–based classification has limited power.
- **Sample design and replication.** scPhylogenomics can be applied to single or multiple tumor samples per patient and across multiple patients. For evolutionary questions (e.g., metastatic seeding), multi-regional or longitudinal sampling greatly increases interpretability. Biological replication at the patient level remains essential for generalizing conclusions. Within each sample, sufficient cell numbers per tumor compartment (typically several thousand) are needed to capture clonal diversity.
- **Controls and orthogonal data.** Where possible, users should complement scRNA-seq with bulk exome or targeted DNA sequencing to confirm key driver mutations and to assess coverage biases in scRNA-derived SNVs. Normal tissues or peripheral blood mononuclear cells can serve as a reference for Panel-of-Normals construction and for benchmarking copykat classifications.
- **Pre-optimization of QC and SNV thresholds.** Ambient RNA, doublet, and mitochondrial filters should be tuned on pilot datasets or subsets of the data, using diagnostic plots to avoid over-filtering (see *Procedure*, Modules 1–2). Similarly, SNV filtering parameters (`min_cells_per_snp`, `min_snps_per_cell`, allele frequency thresholds) may require adjustment based on sequencing depth and chemistry (see *Procedure*, Module 4).

Equipment, expertise, and computational resources

The protocol requires:

- Access to a Linux workstation or HPC cluster with sufficient CPU cores (≥16 recommended), memory (≥64 GB, ideally 128 GB for large projects), and storage (≥100 GB per project).
- Basic familiarity with the Linux command line, container-based execution (Docker or Singularity), and R and Python scripting environments (to run the module scripts described in the *Procedure*).
- Experience with scRNA-seq analysis (e.g., using Seurat) to interpret QC plots, cell-type annotations, and integration results.
- Familiarity with phylogenetic concepts (tree topology, branch lengths, model selection, bootstrap support) to interpret clonal trees and evaluate clustering solutions.

Regulatory approvals

scPhylogenomics is a computational workflow that operates on existing sequencing data and does not involve direct manipulation of human subjects or biological specimens. Regulatory approvals (e.g., institutional review board approval, informed consent, data access agreements) are assumed to have been obtained for the generation and use of the underlying scRNA-seq datasets and are not addressed by this protocol. Users should ensure that appropriate permissions (e.g., data use agreements for controlled-access datasets, Material Transfer Agreements for cell lines) have been secured before downloading or analyzing patient-derived data.

Materials

> **Drafting status:** placeholder. To be expanded in subsequent sessions using the per-module README files in the `scPhylogenomics` GitHub repository (`analyses/<module>/README.md`).

Nature Protocols expects the *Materials* section to enumerate every biological reagent, software dependency, hardware requirement and reference dataset that is required to execute the protocol. For this purely computational workflow, *Materials* will be presented as four subsections: **REAGENTS** (data), **EQUIPMENT** (hardware/software), **REFERENCE DATASETS**, and **REAGENT/EQUIPMENT SETUP**. Pricing/vendor lines are omitted because no wet-lab consumables are used; instead, version pins and DOIs are listed.

REAGENTS (data)

> **TODO** — populate from `scPhylogenomics/README.md` (top-level "Data description" section) and from each module README.

- **Triple-negative breast cancer (TNBC) dataset** — TNBC5 sample from GSE148673 (CopyKAT methodology development study).
- **MSK SPECTRUM HGSOC dataset** — six tumor samples from multiple sites of patient SPECTRUM-OV-003 (`@url:https://www.synapse.org/Synapse:syn25569736/wiki/612269`).
- **Acute myeloid leukemia (AML) dataset** — sample LE1. *(Brief dataset description to be added — see open question SQ-01 in `sessions/`.)*
- **10x Genomics Chromium reference transcriptome** — downloaded from `@url:https://www.10xgenomics.com/support/software/cell-ranger/downloads#reference-downloads`.
- **dbSNP/1000 Genomes common variants** (used by cell-snp-lite Mode 1).
- **REDIportal RNA-editing catalog** (used as optional SNV filter).
- **Panel of Normals (PoN)** — constructed from matched or atlas normal samples (procedure to be documented in the Materials/Procedure cross-ref).

EQUIPMENT (hardware and software)

> **TODO** — pin exact versions used in the case study. Versions should match those in `Dockerfile` at the root of the scPhylogenomics repository.

Category	Tool	Version	Purpose
Hardware	Linux workstation or HPC	—	≥16 CPU cores; 64–128 GB RAM; ≥100 GB storage
Container runtime	Docker	≥24.0	Build & run scPhylogenomics image
Container runtime	Singularity / Apptainer	≥1.2	HPC execution
Aligner	Cell Ranger	TBD	scRNA-seq alignment, demultiplexing
R framework	Seurat	TBD	scRNA-seq object handling, integration
R	SoupX	TBD	

Category	Tool	Version	Purpose
			Ambient RNA correction
R	DoubletFinder	TBD	Doublet detection
R	miQC	TBD	Mitochondrial/ QC filtering
R	SingleR + celldex	TBD	Reference-assisted cell typing
R	copykat	TBD	Ploidy inference
C/Python	cellsnp-lite	TBD	Single-cell SNV calling
Python	SNPmanifold	TBD	VAE-based clonal clustering
Python	NumPy / SciPy / scikit-learn / nimfa	TBD	WNMF clonal clustering
Phylogenetics	FastTree	TBD	Maximum-likelihood trees (fast)
Phylogenetics	IQ-TREE	TBD	Maximum-likelihood trees (model selection, bootstraps)
Visualization	ggplot2 / ggtree / ape	TBD	Tree and clone annotation

REAGENT/EQUIPMENT SETUP

> **TODO** — convert each module README's "Setup" / "Installation" instructions into Nature Protocols numbered Setup paragraphs.

1. **Clone the workflow repository.**

```
git clone https://github.com/ewafula/scPhylogenomics.git
cd scPhylogenomics
```

2. **Pull the prebuilt container** (recommended) **or build from source**:

```
# Singularity (HPC)
singularity pull docker://ghcr.io/ewafula/
        scphylogenomics:latest

# Docker (workstation)
docker pull ghcr.io/ewafula/scphylogenomics:latest
```

3. **Configure the project data tree** under `data/projects/<PROJECT>/...` following the directory layout shown in the top-level README.

4. **Download reference resources** (Cell Ranger reference, common SNP VCF, REDIportal, PoN) into `data/refdata/`.

5. **(Optional) Build a custom reference** for SingleR using `scripts/create-reference-dataset.R`.

Procedure

> **Drafting status:** scaffold. Each module sub-section is a placeholder wrapper. The numbered Nature Protocols steps will be ported in upcoming sessions from `scPhylogenomics/analyses/<module>/README.md` and the wrapper `run-*.sh` scripts. Numbering is contiguous across modules per Nature Protocols house style.

Module 1 — Data preprocessing ● Timing 2–6 h

> Source material: `analyses/data-preprocessing/README.md` and `run-data-preprocessing.sh`.

1. **Prepare Cell Ranger outputs.** *(TODO: paraphrase from Module 1 README, including expected directory layout under `data/projects/<PROJECT>/<SAMPLE>/outs/`.)*

2. **Remove ambient RNA with SoupX.** Run `01-remove-ambient-rna.R`. *(TODO: list parameters, key flags, and expected diagnostic plots.)*

   ▲ **CRITICAL STEP** — Check the SoupX contamination fraction (`rho`) diagnostic plot before continuing.

3. **Detect and remove doublets with DoubletFinder.** Run `02-remove-doublets.R`.

4. **Filter compromised cells with miQC.** Run `03-remove-compromised.R`.

▲ **CRITICAL STEP** — Inspect the per-sample miQC posterior probability plots to ensure that healthy cell populations are retained.

5. **Output check.** Confirm the filtered feature-barcode matrices and sample-level QC tables in `analyses/data-preprocessing/results/`.

Module 2 — Cell typing ● Timing 1–4 h

> Source material: `analyses/cell-typing/README.md` and `run-cell-typing.sh`.

1. **Consolidate samples with `01-consolidate-samples.R`** (Seurat: normalization, optional integration via Harmony / CCA / FastMNN / RPCA, clustering).
2. **Annotate cell types with `02-annotate-cell-types.R`** (SingleR with celldex references, custom Seurat references, or external mapping files).
3. **Output check.** UMAPs, heatmaps, and annotation tables under `analyses/cell-typing/plots/` and `.../results/`.

Module 3 — Ploidy inference ● Timing 1–8 h (CPU-bound)

> Source material: `analyses/ploidy-inference/README.md` and `run-ploidy-inference.sh`.

1. **Classify cell ploidy with `01-classify-cell-ploidy.R` (copykat).**
2. **Inspect copykat heatmaps** and merge ploidy classes into the Seurat metadata.

   ◆ **OPTION** — Skip this module and use external malignancy labels (e.g., from marker-based annotation) when working with near-diploid cancers.

Module 4 — SNV calling ● Timing 4–24 h

> Source material: `analyses/snv-calling/README.md`, `run-snv-calling.sh`, and `utils/run-cellsnp-lite.sh`.

1. **Extract barcodes for the target population** with `01-get-cell-type-barcodes.R`.
2. **Run cellsnp-lite** via `02-snv-calling.py`. *(TODO: enumerate Mode 1 vs Mode 2 parameters, optional REDIportal/PoN filtering.)*

▲ **CRITICAL STEP** — Verify the count of variants surviving filtering (`min_count`, `min_maf`); too few SNVs will starve downstream phylogenetics.

3. **Generate SNP MSAs and mutation profile matrices** with `03-generate-snp-msa.py` (parameters: `min_cells_per_snp`, `min_snps_per_cell`, allele frequency thresholds).

Module 5 — Phylogeny inference ● Timing 4–18 h

> Source material: `analyses/phylogeny-inference/README.md` and `run-phylogeny-inference.sh`.

1. **Infer maximum-likelihood phylogenies** with `01-phylogeny-inference.py` (FastTree or IQ-TREE).
2. **Harmonize variant sets across tree and clustering inputs** with `02-filter-variants.py`.
3. **Cluster clones** with `03-snp-clustering.py` (SNPmanifold VAE *or* hierarchical Ward + WNMF).
4. **Annotate trees** with `04-infer-clonal-phylogeny.R` to overlay clone, ploidy, cell type, and sample of origin and to produce annotated clonal/lineage/sample-level trees.

   ❚ **PAUSE POINT** — Save annotated tree objects (`.rds`/`.nwk`) before downstream DEG/GSEA analyses.

Timing

> **Drafting status:** placeholder. Per-step timings will be filled in after benchmarking on the SPECTRUM-OV-003 case study.

Steps	Description	Approx. time (case study)
1–5	Module 1 — Data preprocessing	2–6 h
6–8	Module 2 — Cell typing	1–4 h
9–10	Module 3 — Ploidy inference (copykat)	1–8 h
11–13	Module 4 — SNV calling (cellsnp-lite + MSA)	4–24 h
14–17	Module 5 — Phylogeny inference	4–18 h

Steps	Description	Approx. time (case study)
	and clonal clustering	
Total	End-to-end execution	**24–48 h** (typical)

> **TODO** — replace approximate ranges with measured wall-clock times on a defined HPC profile (CPU/RAM/threads). Add hands-on vs. compute-only breakdowns where useful.

Troubleshooting

> **Drafting status:** placeholder. Each module README in the scPhylogenomics repository contains a “Troubleshooting / Common issues” block — these will be consolidated here in Nature Protocols house style (Step | Problem | Possible reason | Solution).

Step	Problem	Possible reason	Solution
2	SoupX `rho` cannot be estimated	Too few empty droplets in raw matrix	Provide `tod`/`toc` from raw Cell Ranger output and re-run with manually supplied clusters.
3	DoubletFinder identifies an implausibly large doublet fraction	Heterogeneous tissue with rare populations	Re-estimate `pK` via `paramSweep_v3` and tune the expected doublet rate to the chemistry.
4	miQC drops most cells	Tissue with intrinsically high mitochondrial signal	Replace miQC with a fixed % MT cutoff or run on a tissue-matched calibration dataset.
7	SingleR labels are sparse / “NA”	Reference mismatch	Use a lineage-matched celldex reference or a custom Seurat reference built with `scripts/`

Step	Problem	Possible reason	Solution
			`create-reference-dataset.R`.
9	copykat fails to converge	Too few cells; unclear normal baseline	Provide a normal-cell barcode list explicitly via `norm.cell.names`.
12	cellsnp-lite output is empty	BAM lacks `CB`/`UB` tags or barcodes don't match Cell Ranger output	Re-split BAM with the same barcode list used in Module 1; verify chemistry.
13	SNP MSA collapses to a few cells	`min_cells_per_snp` / `min_snps_per_cell` too strict	Lower thresholds; inspect SNP/cell distribution histograms.
14	IQ-TREE OOM / very slow	Too many sites or cells	Use FastTree first; subset variants by `02-filter-variants.py`; partition by chromosome.
16	SNPmanifold latent space collapses	Insufficient informative SNPs per cell	Reduce VAE latent dim; switch to hierarchical WNMF clustering.

> **TODO** — extend with case-study-specific failure modes encountered during the SPECTRUM-OV-003 / TNBC5 / LE1 runs.

Anticipated results

> **Drafting status:** placeholder. To be expanded once the case-study figures are finalized. Each subsection below corresponds to a planned figure or supplementary item.

Module 1 — Cleaned feature-barcode matrices

Users should obtain per-sample filtered Seurat objects with diagnostic plots showing (i) the SoupX contamination fraction `rho`, (ii) DoubletFinder doublet-score distributions, and (iii) miQC posterior probability surfaces. Typical retention is 60–90% of input barcodes.

> **TODO** — insert representative QC panel for SPECTRUM-OV-003.

Module 2 — Reference-annotated cell atlas

Expected output: an integrated UMAP partitioned into epithelial, stromal, and immune compartments with SingleR labels. For SPECTRUM-OV-003 the case study yields a clear epithelial/aneuploid axis distinguishable from the tumor microenvironment.

Module 3 — Ploidy classification

Expected output: copykat heatmaps showing chromosome-level CNV signal, plus a metadata column partitioning cells into `aneuploid` / `diploid` / `not.defined`. Aneuploid fraction varies by tumor type; HGSOC samples typically exceed 70% aneuploid epithelial cells.

Module 4 — SNP MSAs and mutation matrices

Expected output: a FASTA-style SNP pseudo-MSA (cells × ordered SNP loci) and a binary/tertiary mutation matrix (`0/1/3` for ref/alt/missing). Typical sizes for the case studies range from a few thousand to tens of thousands of SNPs across thousands of cells after default filtering.

Module 5 — Annotated clonal phylogenies

Expected output: maximum-likelihood trees (Newick + plot), clone assignments, and overlay panels showing clone × sample, clone × cell type and clone × ploidy. For SPECTRUM-OV-003 the trees are expected to reveal site-enriched aneuploid clones consistent with intraperitoneal metastasis patterns.

> **TODO** — embed final figures: `images/fig2-cell-typing.png`, `images/fig3-ploidy.png`, `images/fig4-snv-msa.png`, `images/fig5-phylogeny.png`.

Discussion

> **Drafting status:** placeholder. Nature Protocols allows an optional short discussion. Anticipated themes:

- **Strengths.** First end-to-end, containerized scRNA-seq → SNV → phylogeny pipeline; explicit SNP MSAs that integrate with standard phylogenetics software; pluggable clonal clustering.
- **Limitations.** Reliance on expressed loci; aneuploidy-dependence of copykat; computational cost on cohorts with millions of cells.
- **Outlook.** Extension to additional chemistries (Smart-seq, multiome), long-read scRNA-seq, joint inference with copy-number trees, and benchmarking against scDNA-seq–based clonal trees.

Reporting summary

A Nature Research Reporting Summary will be supplied with the final manuscript submission. *(TODO.)*

Data availability

- **TNBC5** scRNA-seq data are available under GEO accession `@url:https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE148673`.
- **MSK SPECTRUM** data are available via Synapse `@url:https://www.synapse.org/Synapse:syn25569736/wiki/612269` (controlled access).
- **AML LE1** data: *TODO — add accession or contact.*

Code availability

The scPhylogenomics workflow is openly developed at `@url:https://github.com/ewafula/scPhylogenomics`. Container images (Docker/Singularity) are released under the same repository. The version used in this protocol is pinned in `build/environment.yml` and in the `Dockerfile`.

Author contributions

> **TODO** — populate using the CRediT taxonomy.

E.W. — software, methodology, writing — original draft. P.D. — methodology, validation. S.M. — conceptualization, supervision, funding acquisition, writing — review & editing.

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Competing interests

The authors declare no competing interests.

References